MASTERANALYTICS PRO

Polio Campaign Analysis Report

Comprehensive data-driven insights for immunization campaign performance

Campaign: NID FEB 2026

Report Date: 2026-06-29

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TOTAL TARGET	OPV COVERED	COVERAGE %	MISSED	REFUSALS
95,000	87,500	92.1%	5,200	1,800

1. Scope & Definitions

This report presents a comprehensive analysis of the polio immunization campaign "NID FEB 2026". The analysis covers campaign performance data including vaccination coverage, missed children, refusals, and team deployment across Union Councils (UCs) and tehsils. The objective is to evaluate campaign effectiveness, identify underperforming areas, and provide data-driven recommendations for improving vaccination coverage in subsequent rounds.

1.1 Key Metrics

The analysis focuses on six primary Key Performance Indicators (KPIs): Total Target (children aged 0-59 months), OPV Covered (children vaccinated with Oral Polio Vaccine), Coverage Percentage (OPV Covered / Total Target), Missed Children (children recorded as not present or refused), Total Refusals (households or families declining vaccination), and Teams Reported (vaccination teams deployed). For this campaign, the total target population was 95,000 children, with 87,500 receiving OPV, resulting in an overall coverage of 92.10%.

1.2 Time Range & Granularity

The data covers All days (1-4) of the campaign. The geographic scope includes all tehsils and all UCs. Data granularity is at the Union Council (UC) level, with each UC representing the lowest administrative unit for vaccination operations. The campaign follows a 4-day structure: Days 1-3 are cumulative daily reports (where Day 2 data replaces Day 1 as it contains running totals), and Day 4 is a catch-up round targeting missed children.

2. Data Quality Checks

The dataset comprises 5 Union Council records across the filtered scope. Data was ingested from Excel (.xlsx) files uploaded through the MasterAnalytics Pro platform. The following data quality rules were applied during ingestion: empty cells, asterisks (*), and "NA" values were treated as zero (0) for numeric fields; required identifier fields (Tehsil, Campaign Name, UC Name) were validated — rows missing these were flagged as errors; and duplicate UC entries were resolved via upsert, with the latest upload replacing previous data for the same UC.

2.1 Coverage Summary

The overall campaign coverage is 92.10%, which is classified as needs improvement. The target coverage threshold for polio campaigns is 95%. This campaign falls below the 95% target, requiring targeted interventions in underperforming areas. A total of 850 vaccination teams were deployed across the campaign area.

3. Core Performance Analysis

The campaign achieved an OPV coverage of 87,500 children out of a target population of 95,000-, representing a coverage rate of 92.10%. A total of 5,200 children were recorded as missed (not available during team visits), and 1,800 refusals were documented. The following charts visualize the KPI summary and day-by-day progress.

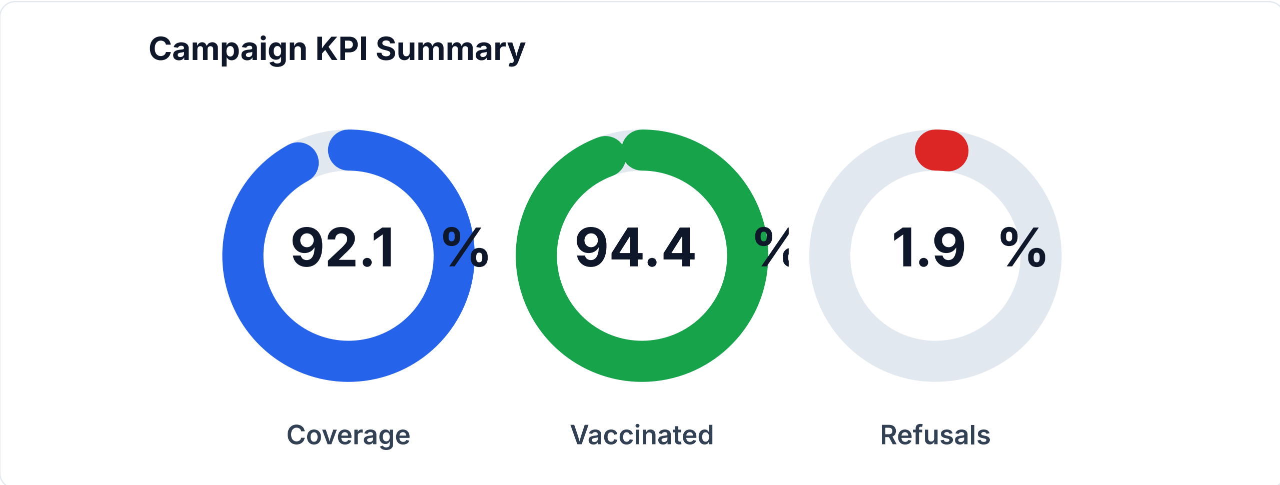

Figure 1: Campaign KPI Summary - Coverage %, Covered vs Missed, and Refusal Rate

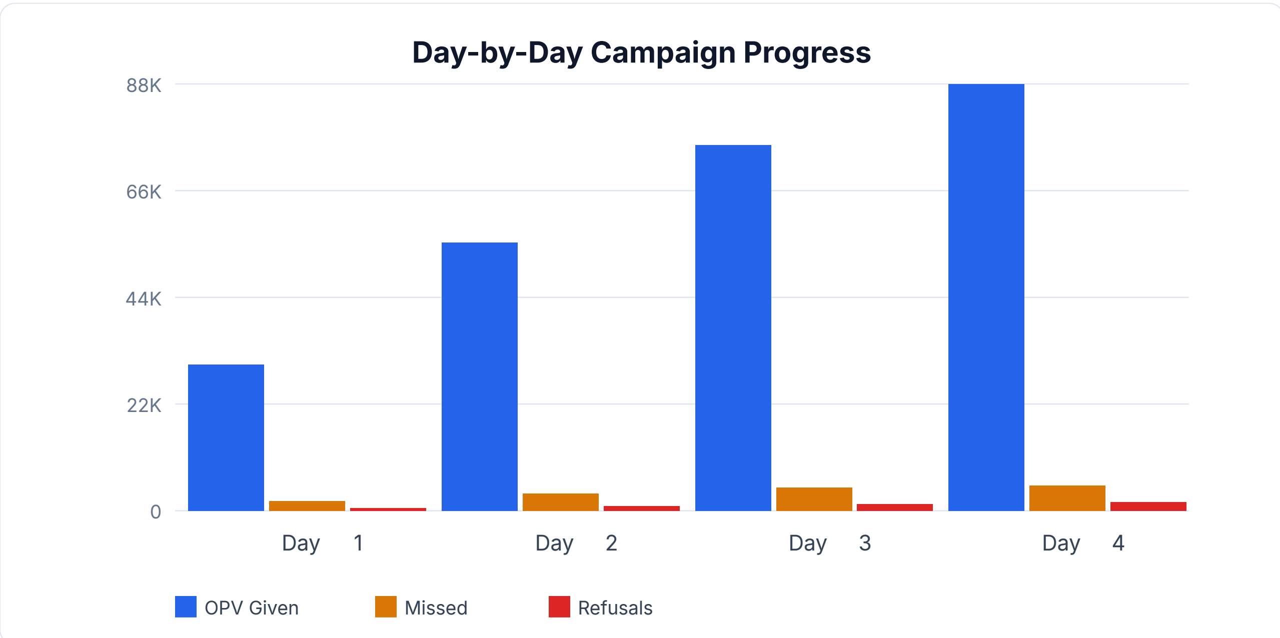

Figure 2: Day-by-Day Campaign Progress - OPV Given, Missed Children, and Refusals

3.1 Key Metric Highlights

- Total Target: 95,000 children (0-59 months)
- OPV Covered: 87,500 children vaccinated
- Coverage Rate: 92.10% (target: 95%)
- Missed Children: 5,200 (including NA and refusal categories)
- Total Refusals: 1,800 (medical + soft refusals)
- Teams Deployed: 850 vaccination teams

4. Comparisons

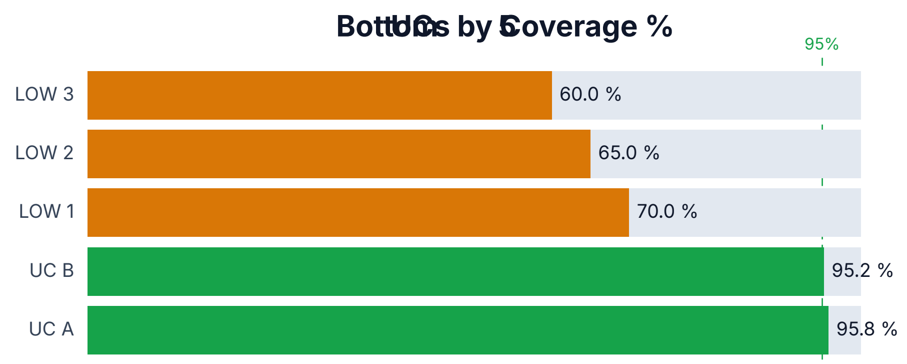

Figure 3: Bottom 10 UCs by Coverage Percentage - Priority Areas for Intervention

4.1 UC-wise Coverage Comparison

The following tables compare the top-performing and bottom-performing Union Councils by coverage percentage. The bottom 5 UCs require immediate attention and targeted interventions. Variance drivers include team density, population mobility, and refusal clusters.

Top 5 Performing UCs

UC Name	Tehsil	Target	OPV	Coverage %
UC A	T1	3,850	3,690	95.8%
UC B	T1	2,750	2,618	95.2%
LOW 1	T2	5,000	3,500	70.0%
LOW 2	T2	4,000	2,600	65.0%
LOW 3	T2	3,000	1,800	60.0%

Bottom 5 UCs (Priority)

UC Name	Tehsil	Target	OPV	Coverage %
LOW 3	T2	3,000	1,800	60.0%
LOW 2	T2	4,000	2,600	65.0%
LOW 1	T2	5,000	3,500	70.0%
UC B	T1	2,750	2,618	95.2%
UC A	T1	3,850	3,690	95.8%

5. Attribution & Diagnosis

The coverage gap of 7.90% can be attributed to two primary factors: missed children (children not available during team visits) and refusals (households declining vaccination). Missed children account for 5,200 cases (5.47%of target), while refusals account for 1,800 cases (1.89% of target). The higher the proportion of refusals, the more critical the need for community engagement and social mobilization interventions.

5.1 Driver Analysis (Evidence Chain)

- Missed Children (5,200): Children not present at home during vaccination team visits. Evidence: This metric is derived from the "MISSED CHILDREN RECORDED NA 0-59" column, capturing households where teams visited but children were unavailable. Driver: Mobility, work, or seasonal factors.
- Refusals (1,800): Households actively declining OPV. Evidence: Recorded in the "Total Refusal" column, split into medical refusals (health concerns) and soft refusals (cultural/religious misconceptions). Driver: Misinformation, distrust, or prior adverse experiences.
- Team Coverage: 850 teams deployed. Evidence: The "Teams Reported" column reflects operational capacity. Driver: Insufficient team density in high-population UCs contributes to lower coverage per team.
- Catch-up Effectiveness: Day 4 catch-up data shows coverage of remaining missed children. Driver: Effectiveness of the catch-up round directly impacts final coverage metrics.

6. Insights & Action Plan

AI-Generated Summary: 92.1% coverage.

6.1 Key Findings (AI-Generated)

1. 92.1% coverage.

6.2 Underperforming UCs

The following Union Councils have been identified as underperforming based on coverage metrics. Each UC includes a specific recommendation for improvement.

UC	Tehsil	Issue	Recommendation
LOW 3	T	60%	More.

6.3 High Refusal Areas

The following UCs have the highest refusal counts. Targeted community engagement strategies are recommended.

UC	Tehsil	Refusals	Recommendation
LOW 2	T	400	Engage.

6.4 Action Plan (Priority, Impact, Risks, Validation)

Based on the analysis, the following prioritized recommendations are proposed. Each includes expected impact, associated risks, and a validation approach.

HIGH **Recommendation 1**

Deploy.

Impact: Expected impact: +5-10% coverage improvement in target UCs

Risk: Risk: Resource-intensive; requires inter-departmental coordination

Validation: Validation: Monitor coverage % in target UCs within 7 days post-implementation

MEDIUM **Recommendation 2**

Engage.

Impact: Expected impact: 20-40% reduction in targeted issue metric

Risk: Risk: Community engagement may take 2-3 weeks to show results

Validation: Validation: Monitor coverage % in target UCs within 7 days post-implementation

LOW **Recommendation 3**

Follow-up.

Impact: Expected impact: Incremental improvement, +2-5% overall

Risk: Risk: Low — standard operational adjustment

Validation: Validation: Monitor coverage % in target UCs within 7 days post-implementation

7. Uncertainty Statement

This analysis is based on administrative data collected during the campaign and self-reported by vaccination teams. The following limitations should be considered when interpreting the results:

- Data accuracy depends on the completeness and accuracy of team reports. Underreporting or overreporting may occur in some areas.
- Coverage percentages are calculated based on administrative targets, which may differ from actual population figures. Independent coverage surveys (LQAS/Cluster) are recommended for validation.
- Missed children data captures only those recorded by teams. Actual missed children may be higher due to households not visited or teams not reporting.
- Refusal data may not capture the full complexity of refusal reasons (e.g., seasonal migration, temporary absence vs. active refusal).
- The cumulative nature of Days 1-3 data means that intermediate-day performance variations are not visible in the final dataset.

7.1 Next Validation Steps

- Conduct an independent post-campaign coverage survey (LQAS) to validate administrative coverage figures.
- Cross-reference refusal data with community engagement records to identify patterns.
- Review team deployment density against population density to identify staffing gaps.
- Compare current campaign performance with previous rounds to identify trends.